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$$f(x) = \ln^2 x$$

$$f'(x) = \frac{2 \ln x}{x}$$

$$\begin{aligned} f''(x) &= 2 \frac{d}{dx} \left(\frac{\ln x}{x} \right) = 2 \frac{d}{dx} \left[\frac{1}{x} \cdot \ln x \right] = 2 \left[\frac{1}{x} \cdot \frac{d}{dx} \ln x + \frac{d}{dx} \left(\frac{1}{x} \right) \cdot \ln x \right] \\ &= 2 \left[\frac{1}{x^2} - \frac{\ln x}{x^2} \right] = 2 \left[\frac{1}{x^2} (1 - \ln x) \right] \end{aligned}$$

$$\begin{aligned} f'''(x) &= 2 \frac{d}{dx} \left[\frac{1}{x^2} (1 - \ln x) \right] = 2 \left[\frac{1}{x^2} \cdot \frac{d}{dx} (1 - \ln x) + \frac{d}{dx} \left(\frac{1}{x^2} \right) \cdot (1 - \ln x) \right] \\ &= 2 \left[-\frac{1}{x^3} - \frac{2}{x^3} + \frac{2 \ln x}{x^3} \right] = \frac{4 \ln x - 6}{x^3} \end{aligned}$$

References